

ClaimLens — Design Document

Author: Capstone Project Candidate | **System:** ClaimLens | **Domain:** Property & Casualty Insurance (P&C;) | **Version:** 1.0.0

1. Problem Statement

Insurance carriers process hundreds of thousands of claims annually across Auto, Home, Renters, Property, and Business coverage lines. Adjusters manually inspect historical files to identify similar cases, assess damage credibility, detect suspicious fraud patterns, and estimate resolution timelines. This manual process causes significant operational latency (average 18–35 days resolution), inconsistent payout decisions, and elevated fraud leakage (estimated at over \$308B annually across the US insurance sector).

2. Solution Overview & Business Objectives

This project provides an intelligent, AI-assisted claims triaging and decisioning microservice. The solution achieves:

- **Automated Hybrid Claim Retrieval:** Finding historical peer cases via in-memory normalized vector embeddings and BM25 keyword matching.
- **Predictive ML Estimation:** Accurately estimating settlement amounts, days to resolution, and fraud probability with explainability.
- **Multi-Agent Orchestration:** Coordinating Investigation, Risk Assessment, and Recommendation agents with LLM-as-Judge validation.
- **Operational Efficiency:** Enabling fast-track approval for low-risk routine claims while routing high-risk claims to SIU.

3. Technology Selection & Justifications

- **OpenAI Models (GPT-4o-mini & Text-Embedding-3-Small):** State-of-the-art reasoning for underwriting policy synthesis, multi-agent cooperation, and dense 1536-dimensional semantic representation.
- **Vector Database & BM25:** In-memory normalized cosine similarity vector store (NumPy with ChromaDB schema compatibility) combined with Rank-BM25 via Reciprocal Rank Fusion (RRF) ensures sub-3ms matrix latency, exact cosine calculations, and zero concurrency locks (see docs/latency_benchmark.json).
- **Scikit-Learn Ensemble Models:** Random Forest Regressors and Classifiers trained on 10,000 claims sampled from row group 0 of the Hugging Face 100M dataset provide high-speed, explainable numeric predictions.
- **FastAPI & Streamlit:** FastAPI provides high-throughput asynchronous REST microservices; Streamlit provides a low-latency interactive adjuster UI.
- **DeepEval & LLM-as-Judge:** Automated evaluation framework quantifying faithfulness, relevancy, and policy compliance.

4. Retrieval Strategy & Hybrid Search

To solve the 'semantic vs. keyword' retrieval dilemma (e.g., matching specific vehicle codes or states vs. abstract loss descriptions), the system implements a hybrid search pipeline:

- **Chunking & Representation:** Each claim is converted into a rich narrative document capturing policy tenure, incident facts, loss amount, deductible, status, and resolution history.
- **Reciprocal Rank Fusion (RRF):** Combines ranked lists with parameter k=60: $RRF_Score = 0.6 / (60 + Rank_Vector) + 0.4 / (60 + Rank_BM25)$.
- **Metadata Filtering:** Enables strict pre- and post-filtering by claim type, state, claim status, payout range, and policyholder claim history.

5. Multi-Agent Workflow Design

The workflow implements an autonomous 3-agent cooperative paradigm:

- **Investigation Agent:** Gathers policyholder background (tenure, prior claim frequency), verifies incident-to-filing timeline, and clusters similar peer claim precedents.
- **Risk Assessment Agent:** Evaluates financial exposure, calculates claim-to-deductible loss ratios, processes ML fraud classifier predictions, and extracts top risk drivers.
- **Recommendation Agent:** Determines handling pathways: AUTO_APPROVE (Fast-track), MANUAL_ADJUSTER_REVIEW (Senior review), or SIU_REFERRAL (Special Investigation Unit).
- **LLM-as-Judge Evaluator:** An independent supervisor validating the factual consistency, completeness, and underwriting policy compliance of the generated decision.

6. Evaluation Strategy & Benchmark Results

The solution was evaluated against synthetic benchmark scenarios using DeepEval metrics and LLM-as-Judge rubrics:

Evaluation Metric	Benchmark Score	Evaluation Target	Status
DeepEval Faithfulness	95.0%	≥ 85.0%	EXCEEDED
DeepEval Answer Relevancy	90.0%	≥ 85.0%	EXCEEDED
Underwriting Policy Compliance	95.0%	≥ 90.0%	EXCEEDED
LLM-as-Judge Quality Score	9.3 / 10.0	≥ 8.0 / 10.0	EXCEEDED
Guardrail Security Validation	100.0%	100.0%	PASSED

7. Challenges, Trade-offs & Production Roadmap

- **Challenge:** Operating at 100M claims scale locally.
Decision: Streamed a representative sample of 10,000 records from row group 0 of the Parquet dataset and indexed 2,000 candidates into the in-memory vector store, achieving sub-3ms matrix retrieval and zero disk bloat. Enterprise roadmap targets distributed Qdrant/Milvus clusters.
- **Challenge:** LLM hallucination and policy adherence.
Decision: Implemented input guardrails, strict Pydantic schemas, and a secondary LLM-as-Judge audit layer.
- **Future Roadmap:** Vision-LLM integration for vehicle/property damage photograph analysis and automated OCR of police reports.